

Colour Space Gamut Mapping Tool

Overview & User Guide

Version 1 · CIE 1931 xy · Polar Plot · Five Mapping Strategies

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Self-contained HTML — no server, no dependencies

1. Overview

The Colour Space Gamut Mapping Tool is a self-contained HTML application for visualising the gamuts of nine industry-standard colour spaces on the CIE 1931 xy chromaticity diagram and a polar plot, and for comparing how five different mapping strategies transform colour data when converting between them. No installation or server is required.

It is designed for photographers, colour scientists, imaging engineers, and colour managers who need to understand exactly what happens to colour data at a gamut boundary — whether converting from ProPhoto to sRGB for web delivery, Rec.2020 to Display P3 for broadcast, or ACES AP0 to ACEScg for VFX.

1.1 Key Capabilities

- Nine colour spaces: sRGB, Adobe RGB, Display P3, Rec.2020, ProPhoto, Hasselblad RGB, Hasselblad Lstar, ACES AP0, ACEScg AP1
- Select any space as source and any other as target; real-time plots and analysis
- Five mapping strategies: Hard Clip, Compress, Hue-Preserve, Chroma Map, Relative Colorimetric
- Bradford chromatic adaptation between D50, D65, and D60
- Quantitative analysis: area %, Δhue° , $\Delta\text{chroma}\%$, $\Delta x\%$, $\Delta y\%$ per primary
- Natural-language primary shift narrative and live cursor coordinates

1.2 Included Colour Spaces

Space	WP	Gamut	Primary Use
sRGB	D65	Narrow	Web, consumer displays
Adobe RGB	D65	Medium	Photography, pre-press
Display P3	D65	Medium-wide	Modern displays, Apple, cinema
Rec.2020	D65	Very wide	UHD broadcast, HDR
ProPhoto	D50	Extremely wide	Raw editing, archival
Hasselblad RGB	D50	Wide	Phocus output
Hasselblad Lstar	D50	Extremely wide	Phocus Lstar output
ACES AP0	D60	Scene-max	VFX mastering
ACEScg AP1	D60	Very wide	CG rendering, compositing

2. Interface

The interface has five zones: title bar, source/target selectors, controls row, plots, and legend/analysis panel. All controls update in real time.

2.1 Source and Target Selectors

Two side-by-side panels each contain a colour-coded button per colour space. The left panel selects the source (space to convert from); the right panel selects the target (space to convert to). Selecting the same space for both hides the analysis panel.

TIP: Source and target can be swapped freely. ProPhoto→sRGB shows compression into a narrow space; sRGB→ProPhoto shows a narrow gamut sitting inside a wide one — useful for understanding containment.

2.2 White Point

D50 (5003 K)	ISO graphic arts standard. Native WP for ProPhoto, Hasselblad spaces. Default.
D65 (6504 K)	Daylight standard. Native WP for sRGB, Adobe RGB, Display P3, Rec.2020.
D60 note	ACES spaces use D60 natively. Bradford matrices adapt D60↔D50 and D60↔D65 automatically.

2.3 View

CIE 1931	Horseshoe diagram, x 0–0.8, y 0–0.9. Spectral locus with wavelength labels. Default view.
Polar	Polar projection centred on active WP. Radius = chroma; angle = hue. Best for comparing hue preservation across strategies.

2.4 On-Canvas Cursor

Hover over the CIE 1931 diagram to read exact xy chromaticity coordinates and the hue angle θ from the white point, displayed in the status bar below the canvas.

3. Mapping Strategies

All five strategies operate in 2D CIE xy chromaticity space, making the mathematics transparent and directly observable on the plot. Each represents a different approach to the gamut boundary problem.

3.1 Hard Clip

Out-of-gamut primaries are moved along the ray from the working white point to the primary, until that ray intersects the target gamut boundary. In-gamut primaries are untouched. Each corner clips independently, so the mapped triangle can become highly asymmetric.

When to use: Hard Clip is the reference baseline showing where each strategy differs. It also models what a naive colour conversion without a colour manager does to out-of-gamut values.

3.2 Compress

Finds the single most out-of-gamut primary and computes a uniform scale factor that brings the entire triangle just inside the target. All three primaries scale by the same factor toward the white point, so the triangle shape is perfectly preserved — only its overall size changes. For very large source gamuts (e.g. ProPhoto vs sRGB) the scale factor can be small (~0.35×), producing a very desaturated result.

When to use: Compress is most meaningful when the source gamut is only slightly larger than the target. For very large sources the compressed triangle will appear extremely small.

3.3 Hue-Preserve

Each source primary is moved to the target boundary along its constant hue angle from the white point (binary search). The hue angle is maintained exactly; only chroma changes. This differs from Hard Clip because the WP ray and the constant-hue ray diverge unless the primary lies exactly radially from the WP.

When to use: Use when maintaining perceived hue is the priority. The polar plot makes hue preservation visible — mapped primaries lie at exactly the same angular positions as source primaries.

3.4 Chroma Map (Perceptual)

Proportional gamut-boundary-to-boundary remapping along each primary hue ray. For each primary: $r_{\text{mapped}} = r_{\text{src}} \times (r_{\text{tgt_boundary}} / r_{\text{src_boundary}})$, where r is the chroma distance from the white point. A primary at 80% of the source boundary maps to

80% of the target boundary. In-gamut primaries also move — scaled proportionally down. Each primary can receive a different scale factor (one per hue direction), unlike Compress which applies a single global factor.

This is the closest xy-space analogue to ICC perceptual rendering intent: it maps the spatial structure of the source gamut onto the target gamut, preserving relative colour relationships.

3.5 Relative Colorimetric

Applies a pure white point translation: the source WP is shifted to the target WP, and all primary xy offsets from their respective WPs are preserved exactly. Algorithm: $\Delta x = \text{primary.x} - \text{srcWP.x}$; $\Delta y = \text{primary.y} - \text{srcWP.y}$; mapped = {tgtWP.x + Δx , tgtWP.y + Δy }. Any mapped primary still outside the target gamut is then hard-clipped.

This models ICC relative colorimetric intent: colours neutral under the source illuminant are reproduced as neutral under the target illuminant. When source and target share the same native WP (both D65), Rel. Colorimetric produces identical results to Hard Clip for in-gamut primaries.

Key insight: Relative Colorimetric is the only strategy that explicitly accounts for WP differences between colour spaces. The other four operate from the single display WP, ignoring native WP differences. Rel. Colorimetric makes the WP shift part of the mapping itself.

3.6 Strategy Comparison

Hard Clip	In-gamut untouched · Out-of-gamut clipped along WP ray · Shape distorts asymmetrically · Ignores native WP diff
Compress	All primaries scaled uniformly · Shape preserved · Driven by worst primary · Ignores native WP diff
Hue-Preserve	In-gamut untouched · Out-of-gamut moved along constant hue ray · Shape differs from clip · Ignores native WP diff
Chroma Map	All primaries remapped proportionally · In-gamut primaries also move · Preserves relative chroma structure · Ignores native WP diff
Rel. Colorimetric	All primaries translated by srcWP→tgtWP · Shape/size preserved relative to WP · Residuals clipped · Accounts for WP diff

4. Reading the Plots

4.1 CIE 1931 xy Diagram

The spectral locus is the horseshoe boundary representing monochromatic light from ~380 nm to 700+ nm. All realisable colours lie within it. Each colour space is a triangle: solid filled = source native; dashed in source colour = mapped source; dashed in target colour = target space; very faint dashes = all other spaces for reference. The white dot marks the active working white point.

TIP: Hover over the CIE 1931 diagram to read exact xy coordinates and hue angle θ from the WP in the status bar below.

4.2 Polar Plot

Radius = chroma (distance from WP); angle = hue. The outer rim = chroma ~0.32 xy units. Particularly useful for comparing strategies: Hue-Preserve and Chroma Map move primaries radially (same angle, different radius); Hard Clip and Compress can also shift hue angle. Switch to polar to see this clearly.

4.3 Plot Colour Rendering

The background is an HSV approximation of CIE colour appearance: hue angle from D65 white point maps to hue; chroma maps to saturation. This is illustrative only, not a calibrated render.

5. Analysis Panel

Appears automatically when source and target differ. Contains a gamut area section and a primary shift table.

5.1 Gamut Area

Three bar charts: Source native % of target area; Target at 100%; Source mapped % after the selected strategy. Narrative sentence explains whether source exceeds or falls short of target, and how much area the strategy retains.

NOTE: Source >100% = the source encodes colours with no equivalent in the target; these must be mapped or clipped. Source <100% = some target colours cannot be reproduced from source data.

Area is computed via the shoelace formula in xy space: $Area = |R_x(G_y - B_y) + G_x(B_y - R_y) + B_x(R_y - G_y)| / 2$. This is a relative indicator, not a perceptual measure (it ignores luminance and perceptual non-linearity).

5.2 Primary Shift Table

Three rows: Target [ref] (reference, no deltas); Source native vs target; Source mapped vs target. The mapped row shows residual error — how close the strategy brings each primary to the target.

xy coords	Chromaticity of the primary at the current display WP
Shift tag	Natural-language direction: "shifted orange/warmer", "shifted cyan/cooler", etc.
Δhue°	Hue angle diff vs target. +ve = clockwise/warmer; -ve = CCW/cooler.
$\Delta chroma\%$	Chroma distance diff, thousandths of xy unit. +ve = more saturated.
$\Delta x\% / \Delta y\%$	Raw xy coordinate differences, thousandths.

5.3 Hue Shift Tags

Warm (orange)	R toward orange; G toward yellow-green
Cool (blue)	R toward purple/magenta; G toward cyan; B toward violet
Green	B toward cyan/green
Neutral	Below significance threshold ($\pm 0.5^\circ$ hue, ± 0.002 chroma)

6. Colour Science Background

6.1 CIE 1931 Chromaticity

CIE 1931 colour matching functions describe the human visual response across the visible spectrum. Integrating a spectral power distribution yields XYZ tristimulus values. Dividing by X+Y+Z gives chromaticity xy — colour direction independent of luminance.

6.2 Bradford Chromatic Adaptation

Colour spaces have different native white points. Converting between them requires adapting all coordinates so the source WP maps to the target WP. This tool uses the Bradford CAT with ICC-standard matrices for D65↔D50 and standard matrices for D60↔D50/D65. The selected display WP controls which adapted version of each primary is shown.

6.3 ICC Rendering Intents vs This Tool

Hard Clip	Analogous to saturation intent: out-of-gamut primaries independently clipped, no spatial adjustment.
Compress	Simplified relative colorimetric with compression: uniform scaling rather than a 3D CMM solution.
Hue-Preserve	Loosely analogous to perceptual intent: hue preserved at expense of chroma.
Chroma Map	Closest analogue to ICC perceptual intent: proportional chroma compression preserving gamut structure.
Rel. Colorimetric	Direct model of ICC relative colorimetric intent: WP-normalised with out-of-gamut clip.

Real ICC CMMs operate in 3D perceptual spaces (CIECAM02, Lab) using lookup tables, not analytical functions. These strategies are geometrically exact in 2D xy but are not direct implementations of ICC rendering intents.

7. Workflow Examples

7.1 ProPhoto to sRGB — Wide Gamut to Delivery

1. Set Source to ProPhoto, Target to sRGB.
2. Area block: ProPhoto covers ~280–300% of sRGB. The scale of compression required is immediately visible.
3. **Hard Clip**: G primary moves most; B barely moves; triangle becomes asymmetric.
4. **Compress**: triangle scales uniformly ($\sim 0.35\times$). Very small mapped triangle due to extreme gamut difference.
5. **Chroma Map**: proportional remap. More even distribution. In-gamut primaries also move inward.
6. **Rel. Colorimetric**: D50 origin shifts to D65; triangle translates; R clips slightly; G clips to sRGB boundary.

7.2 Rec.2020 to Display P3 — Broadcast to Monitor

7. Both spaces use D65 — WP translation has no effect; Rel. Colorimetric = Hard Clip for in-gamut primaries.
8. Rec.2020 B primary is inside Display P3, so only R and G need mapping.
9. Hard Clip vs Hue-Preserve: G clipping point differs — Hue-Preserve preserves hue angle exactly.
10. Use polar view to confirm Hue-Preserve maintains all primaries at the same angular positions.

7.3 sRGB into ProPhoto — Narrow into Wide

11. Set Source to sRGB, Target to ProPhoto. sRGB covers ~33% of ProPhoto in xy area.
12. All five strategies show Mapped ~100% — all sRGB primaries are inside ProPhoto. No mapping needed.
13. The primary table still shows useful data: how far each sRGB primary is from the ProPhoto primary. sRGB G is far from ProPhoto G (ProPhoto G is extreme yellow-green), which explains why raw-to-sRGB conversion loses so much saturated green.

7.4 ACES AP0 to ACEScG AP1

14. Set Source to ACES AP0, Target to ACEScG AP1.
15. AP0 B primary lies far below the spectral locus (virtual primary by design). CIE 1931 view makes this visible.
16. AP0 is designed to encompass all human-visible and camera-capturable colours. AP1 is the working space for CG renders.
17. Hard Clip vs Chroma Map shows the largest contrast here due to the extreme AP0 B primary.

7.5 All Five Strategies — ProPhoto to sRGB

Cycle through all five strategies with ProPhoto source, sRGB target:

- **Hard Clip** — largest mapped triangle; asymmetric; in-gamut primaries fixed
- **Compress** — smallest triangle; perfectly preserves shape; uniform scale
- **Hue-Preserve** — similar to Clip but G/B corners differ; polar shows perfect hue preservation
- **Chroma Map** — between Clip and Compress; all primaries move; relative structure preserved
- **Rel. Colorimetric** — triangle translates by D50→D65 offset, then clips; shape largely preserved

8. Technical Reference

8.1 Bradford Matrices

D65→D50 (ICC):

```
[ 1.0478112,  0.0228866, -0.0501270 ]
[ 0.0295424,  0.9904844, -0.0170491 ]
[-0.0092345,  0.0150436,  0.7521316 ]
```

D60→D50:

```
[ 1.0130,  0.0057, -0.0448 ]
[ 0.0264,  0.9941, -0.0147 ]
[-0.0060,  0.0101,  0.7476 ]
```

D60→D65:

```
[ 0.9871, -0.0061,  0.0576 ]
[ 0.0267,  1.0012,  0.0013 ]
[-0.0012,  0.0000,  0.9991 ]
```

8.2 Colour Space Primaries

Space	WP	R (x,y)	G (x,y)	B (x,y)
sRGB	D65	0.640, 0.330	0.300, 0.600	0.150, 0.060
Adobe RGB	D65	0.640, 0.330	0.210, 0.710	0.150, 0.060
Display P3	D65	0.680, 0.320	0.265, 0.690	0.150, 0.060
Rec.2020	D65	0.708, 0.292	0.170, 0.797	0.131, 0.046
ProPhoto	D50	0.7347, 0.2653	0.1596, 0.8404	0.0366, 0.0001
Hasselblad RGB	D50	0.681, 0.314	0.213, 0.739	0.133, 0.047
Hasselblad Lstar	D50	0.756, 0.244	0.098, 0.902	0.019, 0.022
ACES AP0	D60	0.7347, 0.2653	0.0000, 1.0000	0.0001, -0.0770
ACEScg AP1	D60	0.713, 0.293	0.165, 0.830	0.128, 0.044

Note: ACES AP0 B primary (0.0001, -0.077) is below the line of purples — a virtual primary by design, ensuring AP0 contains all possible camera-captured colours.

8.3 Relative Colorimetric Algorithm

```
// For each primary:
dx = primary.x - srcWP.x;  dy = primary.y - srcWP.y
mapped = { x: tgtWP.x + dx, y: tgtWP.y + dy }
mapped = clipToGamut(mapped, targetGamut, tgtWP)
```

Both srcWP and tgtWP are Bradford-adapted to the display WP before use. The clip step only affects primaries that remain outside the target after translation.

8.4 Adding Colour Spaces

Open the HTML source, locate the `SPACES` object, and add:

```
myspace: { prim: { R:{x:0.64,y:0.33}, G:{x:0.30,y:0.60}, B:{x:0.15,y:0.06} },  
            nativeWP: 'd65', name: 'My Space', color: '#RRGGBB' },
```

The space appears automatically in both selector panels and as a faint reference outline on the plots.

9. Troubleshooting

Analysis panel hidden	Source and target must differ. Same space in both selectors hides the panel.
Compress triangle very small	Expected for large source vs small target. Scale driven by worst primary. Use Chroma Map for a more even result.
Rel. Colorimetric = Hard Clip	Same-WP spaces (both D65) produce identical results for in-gamut primaries — no WP translation to apply.
Primary outside horseshoe	ACES AP0 B and Hasselblad Lstar primaries are intentionally outside the spectral locus — virtual primaries by design.
Polar edges arc near rim	Polar clips triangles to the circular boundary. Extreme primaries appear at the rim; their connecting edges arc.
WP switch changes coordinates	Correct — Bradford adaptation moves all primaries to their positions under the new WP.

10. Quick Reference

Controls

Source panel	Colour space to convert from
Target panel	Colour space to convert to
D50 / D65	Working WP for plot centring and Bradford adaptation
CIE 1931	Horseshoe diagram x 0–0.8, y 0–0.9
Polar	Polar xy centred on active WP
Hard Clip	Per-primary clip along WP ray; in-gamut untouched
Compress	Uniform scale; shape preserved; worst primary sets factor
Hue-Preserve	Move to boundary at constant hue angle from WP
Chroma Map	$r_{\text{mapped}} = r_{\text{src}} \times (r_{\text{tgt_bnd}} / r_{\text{src_bnd}})$; proportional perceptual
Rel. Colorimetric	Translate by $\text{tgtWP} - \text{srcWP}$; clip residuals; models ICC RC intent

Analysis Metrics

Area %	Source area as % of target (xy shoelace)
Δhue°	+ve = clockwise/warmer; –ve = CCW/cooler
$\Delta\text{chroma} \%$	+ve = more saturated; –ve = less saturated
$\Delta x \text{‰} / \Delta y \text{‰}$	Raw xy diff, thousandths
Rows	Target [ref] · Source native · Source mapped — shows residual error after mapping

White Points by Space

D65	sRGB, Adobe RGB, Display P3, Rec.2020
D50	ProPhoto, Hasselblad RGB, Hasselblad Lstar
D60	ACES AP0, ACEScg AP1